

HAZAEEL HANUK ROBSON

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EDUCATION

Vellore Institute of Technology, Vellore, India

Jul 2024 - Present

Master of Science (M.Sc.) in Physics

CGPA: 9.28/10.00

St. Stephen's College, University of Delhi, India

Jun 2023

Bachelor of Science (B.Sc.) in Physical Sciences

CGPA: 8.23/10.00

Doon Heritage School, Siliguri, India

Mar 2020

AISSCE (Class XII) Percentage: 92%

AISSE (Class X) Percentage: 86.2%

Awarded the Merit Scholarship by VIT Vellore for academic excellence during the 1st year M.Sc. Physics

Ranked among the city toppers for 10th and 12th standard by Prabhat Khabar, Siliguri

RESEARCH EXPERIENCE

Stability and symmetry breaking of scalar fields coupled to Gravity

Supervisor: Dr. Soumya Chakrabarti, VIT, Vellore

Jul 2025 - Present

We investigate stability and symmetry breaking for gravity with single and multiple scalar fields in different frames and couplings.

Dynamics of a test charge in Reissner–Nordström spacetime

Supervisor: Dr. Laxmi Narayan Tripathy, VIT, Vellore

Jul 2025 - Oct 2025

Covariant Maxwell equations are solved for a static charge in Reissner–Nordström spacetime under spherical symmetry. Interior/exterior solutions and near-horizon field behavior are analyzed numerically.

Cosmological dynamics of scalar field-coupled interacting dark energy in $f(Q)$ gravity

Supervisor: Prof. P.K. Sahoo, BITS, Pilani, Hyderabad campus

May 2025 - Jul 2025

Interacting dark energy models in symmetric teleparallel gravity are studied using dynamical systems methods to obtain scaling solutions that address the cosmological coincidence problem.

Surface Plasmon Resonance (SPR) based Dual D-shaped Photonic Crystal Fiber (PCF) sensor

Supervisor: Prof. Senthilnathan K, VIT Vellore

Aug 2024 - May 2025

We theoretically designed and optimized an SPR-based dual D-shaped PCF sensor. The sensor detects analytes with refractive indices from 1.33–1.41, covering most biological samples, and shows enhanced wavelength sensitivity through systematically optimized metal thickness and air-hole radii.

RELEVANT COURSEWORK

Physics: Classical Mechanics I, Classical Mechanics II, Electricity and Magnetism, Electromagnetic Theory, Waves and Optics, Solid State Physics, Electronics, Semiconductor Physics, Thermodynamics, Statistical Mechanics, Mathematical Physics, Advanced Mathematical Physics, Quantum Mechanics, Advanced Quantum Mechanics, Advanced Magnetism, Nuclear and Particle Physics, Computational Physics, General Relativity and Cosmology, Astrophysics.

Mathematics: Linear Algebra, Calculus, Group Theory, Real Analysis, Differential Equations, Numerical Methods.

CONFERENCES AND PRESENTATIONS

ICRAPs'26 (International Conference on Recent Advancements in Physical Sciences), National Institute of Technology Uttarakhand, India: 6–8 February 2026 [*Best Paper Presentation Award*]

Presented my work on 'Cosmological dynamics of scalar field-coupled interacting dark energy in $f(Q)$ gravity'. The work was published in the ICRAPs'26 Abstract book.

FAC'26 (National Conference on Frontiers of Astrophysics and Cosmology), St Joseph's University, Bengaluru, India: 19-20 January 2026

Presented my work on 'Stability analysis of scalar field cosmology with spontaneous symmetry breaking'. The work was published in the FAC'26 Abstract book.

LEADERSHIP AND VOLUNTEERING EXPERIENCES

Tutor Oakridge International School, Vishakapatnam)	Dec 2025 – Jan 2026
Taught Standard level IB DP-2 Physics to Hostel students. Guided them on their yearly project.	
Captain Hawabaaz Ultimate Frisbee Club	Jun 2022 – Jul 2023
Player and Captain of the ultimate frisbee team registered under UPAI.	
General Secretary B.Sc. Programme Society, St Stephen's College	Jun 2022 – Jul 2023
Handled finance, designed posts, and maintained event reports.	
NGO Intern Y-Ultimate	Dec 2022 – Feb 2023
Photography, editing, and design for the NGO.	
Photography Intern Sarkari School (NGO)	Dec 2021 – Feb 2022
Worked as a photographer in rural areas of Delhi NCR.	
Executive Council Member Music Society, St Stephen's College	Nov 2021 – Jun 2022
Instrumentalist and council member.	
Tutor Social Service League, St Stephen's College	Jun 2021 - Aug 2021
Taught Physics (Standard 12) in the VidyaJyothi vertical.	
Covid Volunteer St Stephen's College	Mar 2021 – Jul 2021
Volunteered in the food supply wing.	

WORKSHOPS AND CERTIFICATIONS

Gravity as an Effective Field Theory: Jul 2025

1-day workshop by Dr. Claudia de Rham, organized by the National Science and Research Society, Kathmandu.

Dynamical Systems in Cosmology: Jun 2025

2-day workshop organized by the Department of Mathematics, BITS PILANI, Hyderabad Campus.

Greatest Unsolved Mysteries in the Universe: Aug 2022

An online course organized by the Australian National University on the platform Udemy.

Overview of Web GIS Technology: Jun 2021

1-month-long online course organized by the Indian Space Research Organization (ISRO).

SKILLS

Programming: Intermediate knowledge of Python, Fortran, Mathematica.

Database: Intermediate knowledge of MySQL.

Software: Intermediate knowledge of MS Office, COMSOL Multiphysics, ORIGIN.

Languages: English, Hindi, and Malayalam (native).